



ALAGA

3X-MyBIOME Intelligence Ecosystem

A New Standard in Precision Preventive Medicine
Next-Generation Microbiome Smart Wellness



Our Microbial Ecosystem: A Link to Ongoing Health Management

The microbiome interacts with digestion, metabolism and immunity, and changes with our habits.

Approx. **38 Trillion**
**Estimated Bacteria
in the Human Body**

Estimate from a reference adult model; individual variation exists.



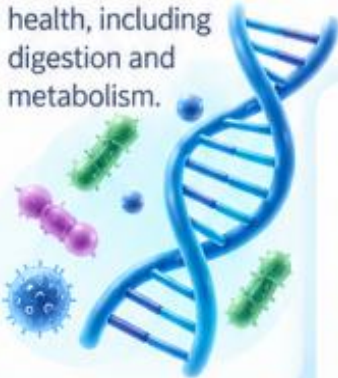
**Interactions
with Immunity**

Gut microbes are closely linked to immune system development and regulation.



**The “Second
Genome”**

Microbial genetic information and functions connect to whole-body health, including digestion and metabolism.



**The Value of
Regular Monitoring**

Repeat testing reveals microbial changes and helps review care plans.



Changes with Lifestyle

Diet, lifestyle habits, medications and other factors influence microbial composition and balance.



Understand Microbial Data → Manage Lifestyle → Track Changes

Beyond the Keyhole: A Broader View of the Microbial Ecosystem

Comparing qPCR for selected targets with NGS for broader community analysis

Targeted Analysis • qPCR



Focus on Selected Targets

**Analysis Scope**

Specific microbial or genetic targets defined by assay design

**Quantification**

Absolute quantification is possible with suitable conditions, such as standard curves

**Key Insights**

Target detection and quantity

Community Analysis • 3BIGS NGS



Explore Diverse Microbes and Community Structure

**Analysis Scope**

Broad microbial communities within the test design

**Quantification**

Primarily relative abundance and community composition

**Key Insights**

Ecosystem information: diversity, balance and dominant microbes

From Target Detection to Community Understanding: **Expand Health Insights with Purpose-Fit Analysis**

Scientific Evidence & Trust: Collaborative Research and Publications

Building expertise through collaboration with research institutions, hospitals and companies, and through academic research.

Key Research Partners & Companies



Korea Disease Control and Prevention Agency



National Cancer Center



Chong Kun Dang Bio



LG Household & Health Care



Ewha Womans University Hospital



Seoul National University School of Dentistry

Selected Research Topics & Publications

01

Investigative and Clinical Urology



Male Urogenital Cancers

Urinary microbiome profiling research

02

Scientific Reports

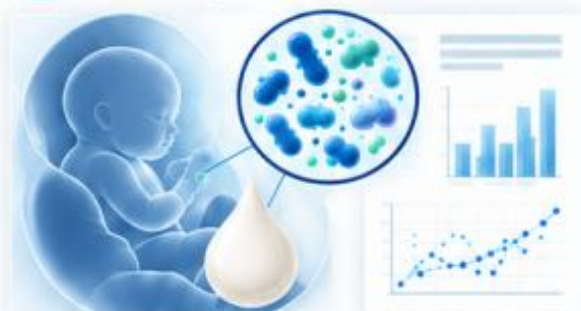


Vaginal Microbial Changes During Pregnancy

Research on characteristics associated with preterm birth

03

Scientific Reports

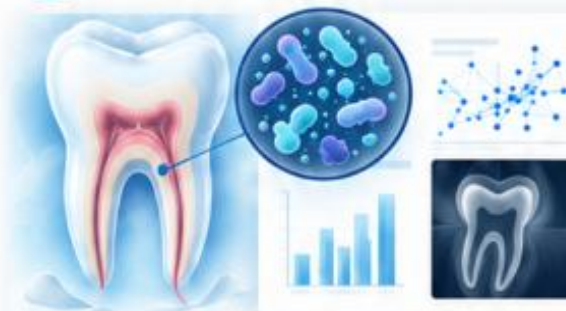


Lactobacillus Distribution During Pregnancy

Research on associations with normal delivery

04

Journal of Endodontics



Persistently Infected Teeth

Microbial biomarker research



Institutional Collaboration +
Academic Research + Data Analysis
to Build Expertise

3BIGS Precisely Analyzes the Microbiome with NGS Technology

From microbial DNA in your sample to community profiles and personalized health insights

4-Step Microbial Analysis Workflow

01

Microbial DNA Extraction

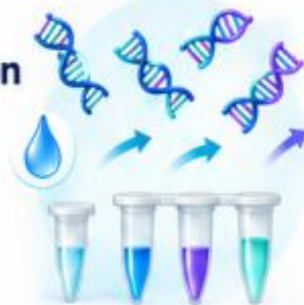
Extract microbial DNA from the sample and check its quality



02

Library Preparation

Prepare libraries for the target analysis, including 16S



03

NGS Sequencing & Analysis

Generate DNA sequences and compare them with databases to analyze microbial taxonomy and composition



04

Interpretation & Care

Interpret community composition, diversity and related indicators to guide care



Integrated Analysis Portfolio



3X-MyBIOME Gut

Gut Ecosystem · Digestive & Metabolic Health



3X-MyBIOME Oral

Oral Ecosystem · Oral Health



3X-MyBIOME Y

Vaginal Ecosystem · CST Classification



DNA
Extraction



Library
Preparation



Community
Analysis



Personalized
Care

Why Choose 3BIGS

Microbiome expertise connecting data, AI, NGS and research partnerships

01 Korean Population Database



Analysis using Korean microbiome data

Interpretation informed by lifestyle and population characteristics

02 AI-Powered Personalized Care



Personal health guidance integrating test and survey data

Tailored action plans and care solutions

03 NGS Microbiome Analysis



Analysis of microbial composition and diversity

Evidence-informed interpretation of results

04 Multi-Site Health Services



An integrated portfolio covering Gut, Oral and Y-Zone

Skin microbiome service in development

05 Research Partnerships



Collaboration through joint research and projects

KDCA, Chong Kun Dang Bio, Ewha Womans University Medical Center, and others

06 Global Data Expansion



Research and analytical data from the Americas, Europe and Asia

Supporting overseas markets and broader result interpretation

**Korean Population Data + AI-Personalized Care
Multi-Site Analysis + Research Partnerships**

The Microbiome: A Starting Point for Whole-Body Health

Our microbial ecosystems interact with immunity, metabolism, women's health and other body functions.

01 Oral Microbiome

Research on links between oral microbes, periodontal health and whole-body health

Dementia Stroke Cirrhosis Diabetes

Associations do not indicate individual disease risk or causation.



02 Gut Microbiome

Close interactions between gut microbes, digestion, metabolism and immunity

Research on associations with obesity, diabetes, skin and digestive health

Digestion Metabolism Immunity



03 Y-Zone Microbiome

The vaginal microbial ecosystem changes across women's life stages

Interpret alongside vaginal, urogenital and pregnancy-related health information

Life Stages Community Balance Women's Health



Integrated Health Management Across Oral, Gut & Y-Zone Microbiomes

3X-MyBIOME Oral

Healthy Change Starts in Your Mouth

Analyze oral microbial imbalance indicators to guide oral and systemic health management.



Turning the Oral Ecosystem into Data

Microbial Analysis from Saliva & Oral Samples

Microbial Composition · Dominant Species · Imbalance Indicators

01 Oral Microbiome Analysis

Analyze microbial distribution and characteristics in saliva and oral samples

Profile the relative composition of diverse oral microbes



02 Oral-Systemic Connections

Interactions between gum inflammation, microbes and bloodstream pathways

Research explores microbial movement through blood and systemic effects



03 Imbalance Insights & Care

Use imbalance indicators to inform oral health management

Guidance for dental consultation, oral hygiene and health management



04 Literature-Informed Interpretation



300+

Reference Publications

Analysis informed by research on oral and systemic health

Journal of Endodontics and more



Oral Microbiome Analysis → Understand Imbalance → Connect Dental & Health Care

3X-MyBIOME Oral Key Features

Systemic Health Links & Oral-Gut Care

Understand oral microbial insights and connect oral and gut health management.



01 Research Links to Systemic Health

Research explores associations between periodontal disease, oral microbes and systemic health.



Dementia

Association
Research



Stroke

Association
Research



Liver Cirrhosis

Association
Research



Diabetes

Association
Research

An association does not establish causation or an individual's disease risk.

02 Integrated Oral-Gut Care **Oral-Gut Axis**

Considering interactions between swallowed oral microbes and the gut ecosystem

An integrated approach to oral hygiene and gut health



03 Personalized Oral Care Solutions



Toothbrushing Routine



Oral Rinsing Habits



Oral Probiotic Product Guide

Personalized care informed by lifestyle habits and product information

Oral Microbiome Analysis → Understand Associations → Personalized Oral-Gut Care

3X-MyBIOME Gut

Gut Microbiome Analysis: A Barometer of Whole-Body Health

Analyze stool microbes to guide digestive, metabolic and immune health management.



Turn the Gut Ecosystem into Data

Approx. 100 Billion Bacteria per Gram of Stool

Bacterial counts vary by sample and individual.

01 Why the Gut Ecosystem Matters

Interactions between gut microbes, immunity and metabolism

Research on links with metabolic health, including obesity and diabetes.



02 Next-Generation Sequencing (NGS)

Microbial taxonomy and composition analysis using 16S ONT sequencing

Explore the distribution and characteristics of diverse gut microbes.



Gut Microbiome Dietary Types

Example Service Categories



Fat Tank

Traits related to fat and protein intake



Vegetable Enthusiast

Traits related to dietary fiber utilization



Sugar Lover

Traits related to carbohydrate intake and metabolism

These are service categories, not criteria for diagnosing obesity or other diseases.

Digestive Health Self-Check

Using the Bristol Stool Scale

- ☒ Bowel movement frequency
- ☒ Stool form
- ☒ Discomfort during bowel movements
- ☒ Digestive symptoms



Linked to a Database of 450+ References



Microbial Composition + Lifestyle Data + Digestive Health Indicators for Personalized Care

3X-MyBIOME Gut

Key Features & Health Indicator Screening

Microbial insights across 14 health categories with practical guidance for personalized care.



From Analysis to Personalized Care

Microbial Composition >

Related Indicators >

Lifestyle Care

01 Indicators Across 14 Health Categories

Selected Examples



General & Metabolic Health

Obesity · Diabetes · Hypertension



Skin Health

Atopic Dermatitis · Psoriasis



Digestive & Cancer Research Indicators

Stomach Cancer · Colorectal Cancer · IBS



Good



Moderate



Caution

Example microbial indicators ·
Not a diagnosis

02 Tracking Key Microbial Profiles



Akkermansia muciniphila

Research on metabolic health associations



Staphylococcus aureus

Information on potential pathogenicity

Detection · Relative Abundance (%) · Change Over Time

03 Solutions for Healthier Habits

Personalized care for individual needs



Bristol Stool Scale-based dietary guidance



Product selection guidance with disclosed strains and CFU

Understand Health Insights → Track Microbial Changes → Diet & Product Guidance

3X-MyBIOME Y

Vaginal Microbiome Analysis for Women's Health

Analyze microbial changes across life stages and classify communities using the CST model.



Turn the Vaginal Microbiome into Data

Microbial Composition • Distribution • Community Types

Considering life stages and lifestyle together

01 Vaginal Microbiome Analysis

Analyze microbial distribution and Lactobacillus dominance.

Composition

Dominant
Microbes

Community
Diversity

02 CST (Community State Types) Classification



CST I

L. crispatus

Dominant

Associated with a protective environment.



CST II

L. gasseri

Dominant

Lactobacillus-dominant



CST III

L. iners

Dominant

Interpret alongside context and symptoms.



CST IV

Mixed Non-Lactobacillus Community

Associated with diversity, inflammation and imbalance.



CST V

L. jensenii

Dominant

Lactobacillus-dominant



ETC

Other Mixed Types

Additional service categories.

CST classifies microbial communities and does not diagnose disease on its own.

03 Interpretation Based on Literature & Clinical Guidelines



250+
References

PNAS

CDC

ACOG

Results explained with reference to research literature and clinical guidelines.

Life Stage + Microbial Composition + CST Type for Personalized Care

3X-MyBIOME Y Key Features

18 Health-Related Items & Longitudinal Monitoring

Visualize five health domains and track microbial changes through repeat testing.



01 5 Health Domains • 18 Reference Items

● Good ● Caution

	Basic Health	Vaginal odor • Vaginal dryness • Vulvar pain/itching • Vaginal discharge	● ● ● ● ●
	Vaginal Health	Bacterial vaginosis • Aerobic vaginitis • Trichomoniasis • Chlamydia	● ● ● ● ●
	Urogenital Health	Pelvic inflammatory disease • Cystitis • Non-gonococcal urethritis	● ● ● ● ●
	Uterine Health	Research on cervical cancer • Endometrial cancer • Endometriosis	● ● ● ● ●
	Pregnancy Health	Research on preterm birth • Infertility • Gestational diabetes • Preeclampsia	● ● ● ● ●

Items support result explanations and consultations; dots illustrate microbial indicators. They do not confirm or predict infections, cancer or pregnancy complications.

02 Longitudinal Tracking

Track changes in the relative abundance of beneficial and monitored microbes.



03 Personalized Care for Women



Vulvar Cleansing & Lifestyle Guidance

Product guidance based on ingredients and strains.

L. rhamnosus GR-1

L. reuteri RC-14

Example strains; review individual circumstances and product information together.

Understand Health Data → Track Microbial Changes → Personalize Care

One-Stop Testing at Home or Work

From booking to results with the ALAGA mobile healthcare app



01

Order a Package

Book and pay in the app



02

Kit Delivery

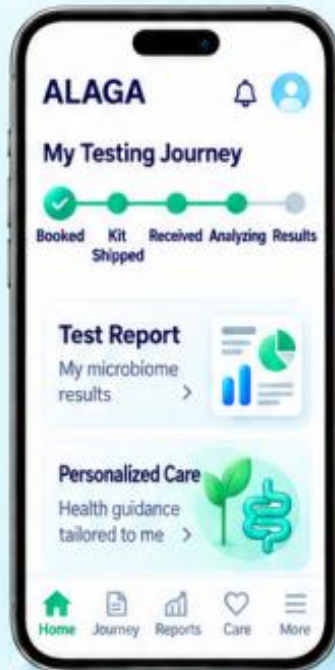
Receive your test kit at home or work



03

Easy 3-Minute Sampling

Collect your sample and return it



ALAGA means care and nurturing



04

Laboratory QC

Check sample quality



05

NGS Analysis

Analyze microbial taxonomy and composition



06

Smart Results

View results and care guidance in the app

Order → Delivery → Self-Sampling → QC → NGS Analysis → Results

ALAGA Testing: Collaborating Research Institutions



Seoul National University



Yonsei University



Korea University

KAIST

POSTECH

UNIST

NGS Analysis Process | Four steps to scientific microbiome analysis



01

DNA Extraction

Extract microbial DNA from the sample



02

Library Preparation

Prepare DNA libraries



03

Sequencing & Analysis

Analyze DNA sequences and microbial taxonomy with NGS



04

Interpretation & Care

Provide personalized health guidance

ALAGA Test Portfolio | Key testing services at a glance



Gut Microbiome Test

Gut health supports whole-body health



Oral Microbiome Test

Oral health connects to whole-body health



Y Microbiome Test

Supporting immune balance and everyday wellness

ALAGA Mobile App & 7-Step Journey

An intuitive one-stop healthcare app built around care and nurturing



7-Step End-to-End User Journey



6-Stage Real-Time Tracking



ALAGA
From Booking to Ongoing Care



Barcode-Based 1:1 Mapping

Scan the kit barcode to link the participant identification code



Improve Sample Identification Accuracy · Prevent Labeling Errors

Booking · Testing · Results · Care · Tracking in One App

Intuitive Reports & Detailed Screening Across 5 Health Domains

Simplify complex microbiome data and view health reference indicators at a glance

01 CST Classification

Visualize vaginal community types and representative beneficial bacteria



Classification Based on CST (Community State Types)

02 5 Health Domains · 18 Detailed Indicators

Good/Caution Dot Gauges by Domain

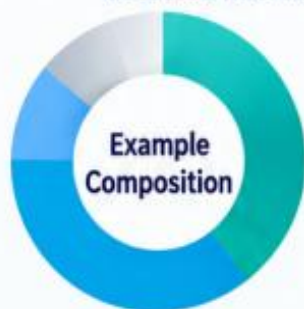
● Good ● Caution

	Basic Health	Information related to odor and itching	● ● ● ● ●
	Vaginal Health	Microbial information related to vaginitis	● ● ● ● ●
	Urogenital Health	Urinary and reproductive health references	● ● ● ● ●
	Uterine Health	References for uterine health consultations	● ● ● ● ●
	Pregnancy Health	References for pregnancy and reproductive health consultations	● ● ● ● ●

Example display; actual results vary.

03 Microbial Composition Donut Chart

Relative proportions (%) of the most abundant detected microbes



- Lactobacillus
- Prevotella
- Other

Relative abundance differs from absolute bacterial counts.



04 Detailed PDF for Clinical Consultations



48~50 Pages

Download the Full Report Instantly in the App

Use test results to support consultations and review care plans

Clear Summary + Detailed PDF to Support Understanding & Consultation

Longitudinal Tracking & Alaga AI Doctor

Track health changes through repeat testing and understand results through AI conversations.



01 Longitudinal Comparison



CST Changes



Beneficial Bacteria
Trends



Compare Past Results



Example Comparison View



02 Alaga AI Doctor

24/7 Conversational AI Results Guidance



Test Date



CST Results



Detected
Microbes



Personalized Guidance Linked to Your Latest Test Data

Understand Results • Explain Care Plans • Retesting Guidance



AI Summary

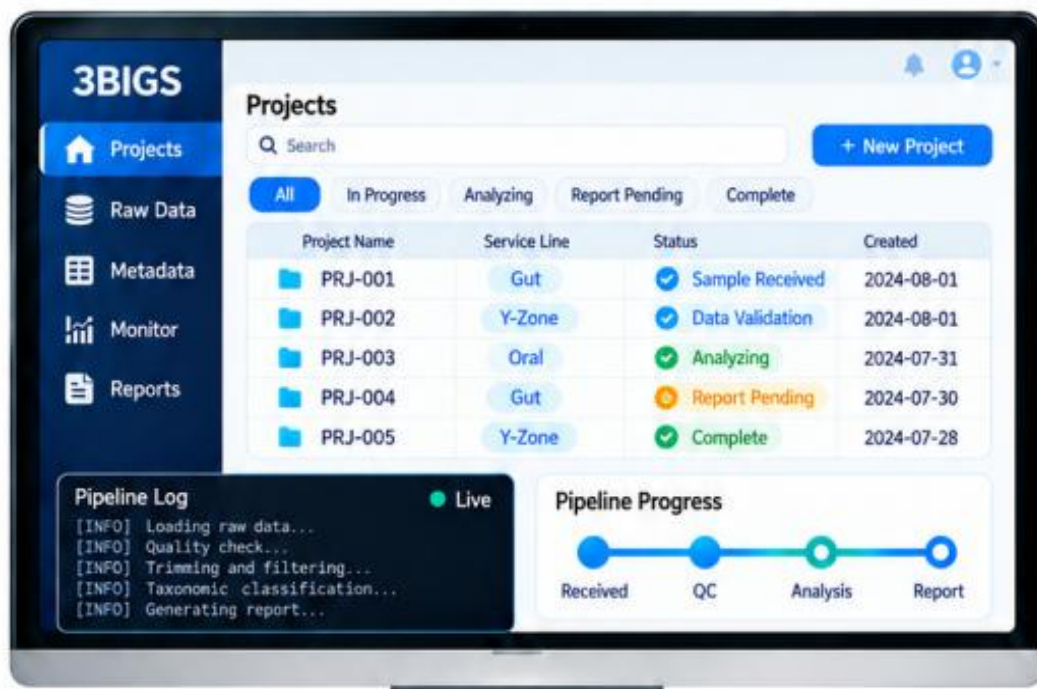
Clearly summarize indicators that changed since the previous test

Review Changes → Check Care Plans → Regular Retesting

Track Result Changes + Ask AI Questions Anytime

3BIGS Integrated Admin Platform & 5-Step Analysis Pipeline

A one-stop B2B console connecting sample receipt, QC, analysis and report issuance



Integrated Operations Workspace



Real-Time Monitoring



Admin Approval

01



Create Project

Generate a project ID and select a service line

Gut Y-Zone Oral



02



Raw Data

Upload 16S ONT NGS sequencing files

Fastq.gz · Automatic File Validation



03

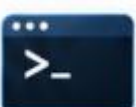


Build Metadata

Map participant ID, sex, age and collection date



04



Run & Monitor

View real-time logs and pipeline progress (%)

Support early error detection and response



05



Report Center

Automatically generate Korean and English (KR+EN) PDFs; issue after admin approval



Create Project → Validate Data → Build Metadata → Monitor Analysis → Issue Report

Lab/QC Kanban Board & Results AI

Review and Approval Engine

Results Engine

Strict QC and multi-stage review support analysis quality and safe results release

01 Real-Time Lab/QC Kanban Board

6 Status Lanes · Drag & Drop

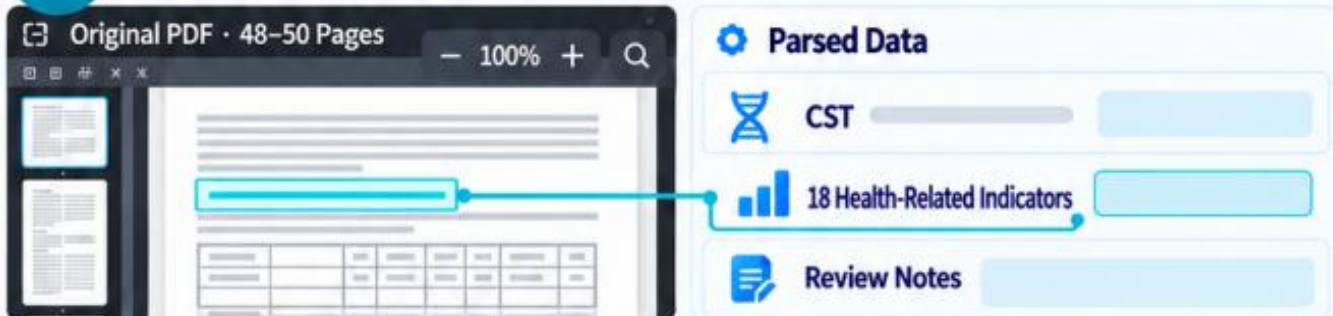


Screen samples against quality criteria, including contamination and low concentration

QC Passed → Analysis | QC Failed → Review Resampling

02 Results Engine

Split View · 1:1 Comparison



Compare original and parsed data by section · Manual Corrections · Review Notes

03 Multi-Stage Approval Before Customer Release



Sample QC → Review Against Original → Final Approval → Customer Release

Evidence-Based Personalized Solutions & Commerce Integration

Connect test results to actionable guidance and informed product choices



From Analysis to Action

Action Guidelines

Product Guide

01

Care Rule Engine

Analyze test values and survey data using 37 rules

TOP 3 Personalized Guidelines Without Duplication



Tests + Surveys

Apply Rules

Action Guidelines

02

Probiotic Guidance Based on Disclosed Ingredients & Strains

Transparent product information on strains, CFU and ingredients

L. rhamnosus GR-1

L. reuteri RC-14

Example strains; review individual circumstances and product information together



03

Lifestyle Curation

Linked to a library of 84 expert solutions



Daily Care



Sleep Management



Y-Zone Cleansing Habits



Tests & Surveys → TOP 3 Guidelines → Products & Lifestyle Care

104 Entities: Data Ecosystem & Security

ALAGA Canonical Schema v1 structures biological big data and clinical knowledge

ALAGA Canonical Schema v1



14
Domains

104
Entities



Integrated Data Governance

01

Integrated Domain & Entity Database

Key Domains



Biological
Knowledge
18



Clinical
Knowledge
12



Scientific
Evidence
14



Products &
Solutions
12



Tests &
Measurements
10



Personal
Health Data
14



AI &
Intelligence
7

7 Key Domains: 87 Entities • All 14 Domains: 104 Entities

02

300+ Research Publications & Guidelines

Link research literature and clinical guidelines 1:1 with knowledge items



CDC

ACOG

PNAS

Example Literature & Guideline Sources

03

Role-Based Access Control & Audit Logs

RBAC • Audit Log



Super Admin

Lab

Ops

Content Manager

Separate permissions by role and retain
access and change histories



Accountability &
Transparent Data Management

Structured Data + Traceable Evidence + Role-Based Security

ALAGA Database Overview

An integrated data ecosystem connecting 104 entities, from microbial knowledge to personalized care

01

Biological Knowledge

Approx. 18 Entities



- Taxon • Organism • Species • Strain
- Function • Metabolite • Body Site

Microbial Taxonomy • Functions • Body Sites

02

Clinical Knowledge

Approx. 12 Entities



- Health Condition • CST Reference
- Clinical Feature • Epidemiology
- Clinical Guideline • Condition Relationship

Health Conditions • Clinical Features • Guidelines

03

Scientific Evidence

Approx. 14 Entities



- Source • Publication • Study
- Clinical Evidence • Evidence Link
- Population • Outcome

Literature • Research • Evidence Links

04

Products & Solutions

Approx. 12 Entities



- Test Product • Health Product
- Formulation • Ingredient
- Product Claim • Care Solution

Products • Strains • Ingredients • Care Information

05

Test & Measurement

Approx. 10 Entities



- Test Definition • Test Instance
- Specimen • Analysis • Measurement
- Result • Report

Specimens • Analysis • Results • Reports

06

Personal Health

Approx. 14 Entities



- User • Profile • Consent
- Test History • Result History
- Personal Record • Subscription

Consent • Test History • Personal Records

07

AI & Intelligence

Approx. 7 Entities



- AI Knowledge • Conversation • Message
- Context Reference • Response Provenance
- AI Safety • AI Evaluation

Conversations • Response Evidence • Safety Evaluation

08

Governance & System

Approx. 16 Entities



- Reference Data • Version • Data Quality
- Import Batch • Audit Log
- Admin Preference • System Config

Versions • Quality • Audits • System Settings